ELITE

IGCSE Mathematics A • Pearson Edexcel

14_November_2025_provisional • Unit 1

1H + 2H source-order past paper

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REVIEWED PROVISIONAL | L-to-M

Converted from Linear Higher papers; not an official Pearson paper or mark scheme.

Contents

2 By rounding each number to one significant figure, work out an estimate for the value of

$$\frac{2.11^2 \times 58.9}{\sqrt{8.859}}$$

Show your working clearly.

....

(Total for Question 2 is 2 marks)

ANSWER / WORKED SOLUTION

1H • Family: Estimate a numerical calculation • Variant: One-significant-figure estimate • USER-ATTESTED LEGACY SOLUTION

Topic check: Rounding, Estimation and Bounds. The tag is correct.

Method

Round each number to one significant figure:

2.11approx 2,quad 58.9approx 60,quad 8.859approx 9

(2.11^2× 58.9)/(√(8.859)) approx (2^2× 60)/(sqrt 9)

=(4× 60)/(3)=80

Answer: 80

7 The weight of a bag of apples is 475 g correct to the nearest g

(a) Write down the lower bound of the weight.

.... g
(1)

The height of a box is 120 cm correct to the nearest 10 cm

(b) Write down the upper bound for the height.

.... cm
(1)

(Total for Question 7 is 2 marks)

ANSWER / WORKED SOLUTION

1H • Family: Recover single-measurement bounds • Variant: Mixed stated accuracies • USER-ATTESTED LEGACY SOLUTION

Topic check: Rounding, Estimation and Bounds. The tag is correct.

Method

The weight 475 g correct to the nearest g has lower bound

$475-0.5=474.5$

The height 120 cm correct to the nearest 10 cm has upper bound

$120+5=125$

Answer: (a) 474.5 g, (b) 125 cm

8 $\dfrac{8^{-2} \times 8^{9}}{8^{10}} = 8^{n}$

Work out the value of n

$n =$ ....

(Total for Question 8 is 2 marks)

ANSWER / WORKED SOLUTION

1H • Family: Combine indexed algebraic factors • Variant: Multiply or divide common bases • USER-ATTESTED LEGACY SOLUTION

Topic check: Algebraic roots and indices. The tag is correct.

Solution

$\frac{8^{-2}\times 8^{9}}{8^{10}}=8^{-2+9-10}=8^{-3}$

Answer: $n=-3$

10 (a) Simplify $(6m)^0$

....

(1)

(b) Solve $5y + 20 < 7y + 1$

....

(2)

(c) Factorise fully $15w^2x^5 + 25w^3x^2$

....

(2)

(d) Simplify fully $\dfrac{36a^5c^7}{12a^2c^3}$

....

(2)

(Total for Question 10 is 7 marks)

ANSWER / WORKED SOLUTION

1H • Family: Extract an algebraic common factor • Variant: Common factor followed by a second factorisation • USER-ATTESTED LEGACY SOLUTION

Topic check: Factorising. The tag is acceptable for this mixed algebra question.

Solution

(a) $(6m)^0=1$.

(b)

$5y+20<7y+1$

$19<2y$

$y>(19)/(2)$

(c)

$15w^2x^5+25w^3x^2=5w^2x^2(3x^3+5w)$

(d)

$(36a^5c^7)/(12a^2c^3)=3a^3c^4$

Answer: 1, $y>(19)/(2)$, $5w^2x^2(3x^3+5w)$, $3a^3c^4$

11 *ABC* is a right-angled triangle.

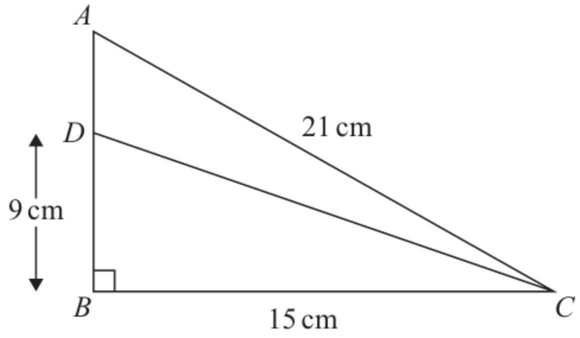


Diagram **NOT** accurately drawn

$AC = 21$ cm $\quad BC = 15$ cm $\quad$ angle $ABC = 90°$

The point D lies on AB such that $DB = 9$ cm

Work out the size of angle ACD
Give your answer correct to one decimal place.

.... °

(Total for Question 11 is 4 marks)

ANSWER / WORKED SOLUTION

1H • Family: Use angles of elevation and depression • Variant: Find a height from an elevation or depression angle • USER-ATTESTED LEGACY SOLUTION

Topic check: Right-Angled Triangles - Pythagoras & Trigonometry. The tag is correct.

Solution

First find AB:

AB^2=21^2-15^2=216

AB=14.696ldots

AD=AB-DB=14.696ldots-9=5.696ldots

Angle ACB:

cos ACB=(15)/(21)

ACB=44.415ldots^circ

Angle DCB:

tan DCB=(9)/(15)

DCB=30.963ldots^circ

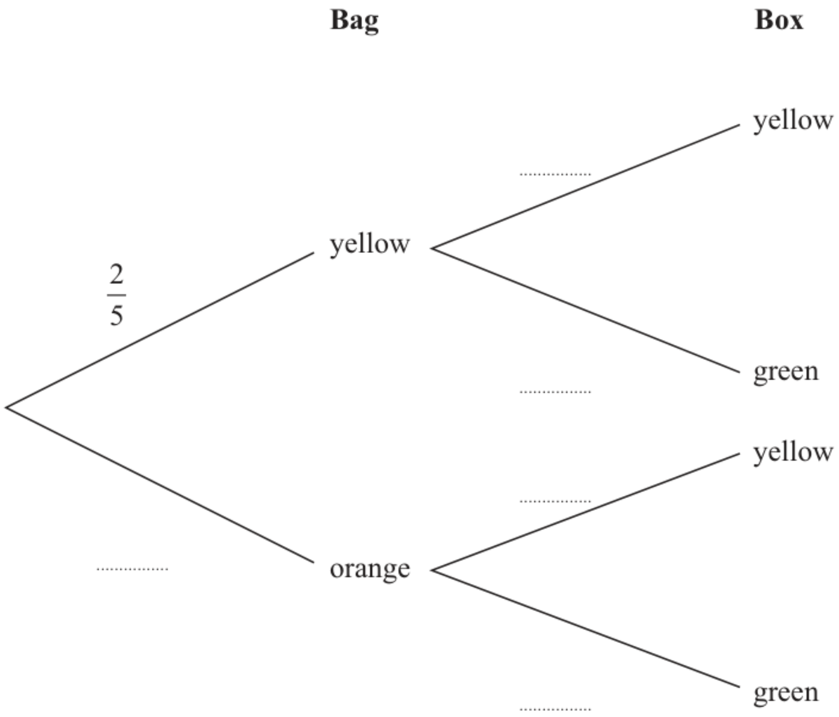
ACD=44.415ldots-30.963ldots=13.451ldots^circ

Answer: 13.5^circ.

13 In a bag, there are only 2 yellow beads and 3 orange beads.
In a box, there are only 3 yellow beads and 5 green beads.

Chi takes at random a bead from the bag and a bead from the box.

(a) Complete the probability tree diagram.



(2)

(b) Work out the probability that Chi takes two beads of different colours.

....

(3)

(Total for Question 13 is 5 marks)

ANSWER / WORKED SOLUTION

1H • Family: Represent and read events with Venn diagrams • Variant: Complete a two-set Venn diagram from totals • USER-ATTESTED LEGACY SOLUTION

Topic check: Probability Diagrams - Venn & Tree Diagrams. The tag is correct.

Solution

Bag 1:

P(Y)=frac25,qquad P(O)=frac35

Bag 2:

P(Y)=frac38,qquad P(G)=frac58

The only way to get the same colour is yellow then yellow.

P(different)=1-frac25×frac38

=1-(3)/(20)=(17)/(20)

Answer: (17)/(20).

14 (a) Expand and simplify $7x(3x + 2)(2x - 5)$

....

(3)

(b) Solve $\dfrac{9}{2y} + \dfrac{5}{7} = 5$

Show clear algebraic working.

$y =$ ....

(3)

(Total for Question 14 is 6 marks)

ANSWER / WORKED SOLUTION

1H • Family: Distribute a single term • Variant: Positive monomial over a bracket • USER-ATTESTED LEGACY SOLUTION

Topic check: Expanding brackets. The tag is acceptable.

Solution

(a)

$7x(3x+2)(2x-5)=7x(6x^2-11x-10)$

$=42x^3-77x^2-70x$

(b)

$\frac{9}{2y}+\frac{5}{7}=5$

$\frac{9}{2y}=\frac{30}{7}$

$63=60y$

Answer: (a) $42x^3-77x^2-70x$, (b) $y=\frac{21}{20}$

15 AOB is a sector of a circle with centre O and radius 12 cm

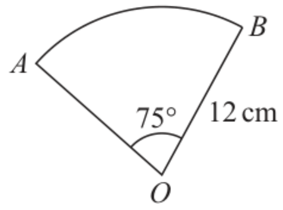


Diagram **NOT** accurately drawn

Work out the area of the sector.
Give your answer correct to 3 significant figures.

.... cm^2

(Total for Question 15 is 2 marks)

ANSWER / WORKED SOLUTION

1H • Family: Use arc and sector measures • Variant: Recover radius or central angle from an arc/sector measure • USER-ATTESTED LEGACY SOLUTION

Topic check: Circles, Arcs & Sectors. The tag is correct.

Solution

area=(75)/(360)× π(12)^2

=94.247ldots

Answer: 94.2 cm^2.

19 The diagram shows triangle *ABC*

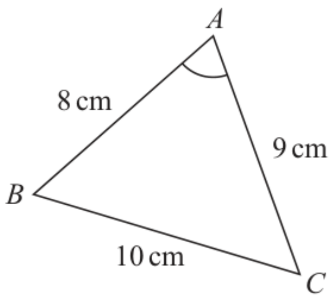


Diagram **NOT** accurately drawn

$AB = 8\text{ cm}$ $\quad BC = 10\text{ cm}$ $\quad CA = 9\text{ cm}$

Work out the size of angle *BAC*
Give your answer correct to one decimal place.

.... °

(Total for Question 19 is 3 marks)

1H • Family: Use area equals one-half ab sine C • Variant: Recover a side or angle from triangle area •
USER-ATTESTED LEGACY SOLUTION

Topic check: Corrected to Sine, Cosine Rule & Area of Triangles.

Solution

Use the cosine rule for angle BAC:

$\cos A=\frac{8^2+9^2-10^2}{2(8)(9)}$

$\cos A=\frac{45}{144}$

$A=71.790\ldots^\circ$

Answer: 71.8°.

20 The straight line **L** has equation $y = 4x + 7$

The straight line **M** is perpendicular to **L** and passes through the point with coordinates (8, 1)

Find an equation for **M**
Give your answer in the form $y = mx + c$

....

(Total for Question 20 is 3 marks)

ANSWER / WORKED SOLUTION

1H • Family: Read and plot Cartesian coordinates • Variant: Read a plotted coordinate pair • USER-ATTESTED LEGACY SOLUTION

Topic check: Correct.

Method

Line L has equation

y=4x+7

So its gradient is

4

Line M is perpendicular to L, so its gradient is

-frac14

Line M passes through (8,1).

y-1=-frac14(x-8)

y-1=-frac14x+2

y=-frac14x+3

Answer: displaystyle y=-frac14x+3

21 Express $5x^2 - 20x + 23$ in the form $a(x - b)^2 + c$ where a, b and c are integers.

....

(Total for Question 21 is 3 marks)

ANSWER / WORKED SOLUTION

1H • Family: Complete the square for a monic quadratic • Variant: Monic quadratic x^2+bx+c • USER-ATTESTED LEGACY SOLUTION

Topic check: Correct.

Method

5x^2-20x+23=5(x^2-4x)+23

=5((x-2)^2-4)+23

=5(x-2)^2-20+23

=5(x-2)^2+3

Answer: 5(x-2)^2+3

22 There are 15 buttons in a box.

3 of the buttons are red
2 of the buttons are pink
10 of the buttons are blue

Pete takes at random three buttons from the box.

Work out the probability that there is still at least one pink button in the box.

....

(Total for Question 22 is 3 marks)

ANSWER / WORKED SOLUTION

1H • Family: Combine selections without replacement • Variant: Calculate one ordered no-replacement path
• USER-ATTESTED LEGACY SOLUTION

Topic check: Correct.

Method

There are 2 pink buttons.

At least one pink button is still in the box unless both pink buttons are taken.

So use the complement.

The probability that both pink buttons are taken is

frac{binom{2}{2}binom{13}{1}}{binom{15}{3}}

=(13)/(455)

=(1)/(35)

Therefore,

P(at least one pink is still in the box) =1-(1)/(35)

=(34)/(35)

Answer: (34)/(35)

23 $a = \dfrac{2x + 5}{1 - x}$ $\quad x = \dfrac{5 - 2y}{3y}$

Write a in the form $\dfrac{m + ny}{p(y - 1)}$ where m, n and p are integers.

Show your working clearly.

$a =$ ....

(Total for Question 23 is 3 marks)

1H • Family: Substitute or transform a rational expression • Variant: Substitute a variable relation into a fraction • USER-ATTESTED LEGACY SOLUTION

Topic check: Corrected from Ratio Toolkit to Algebraic Fractions.

Method

$a=(2x+5)/(1-x),quad x=(5-2y)/(3y)$

Substitute for x.

Numerator:

$2x+5=2((5-2y)/(3y))+5$

$=(10-4y+15y)/(3y) =(10+11y)/(3y)$

Denominator:

$1-x=1-(5-2y)/(3y)$

$=(3y-5+2y)/(3y) =(5y-5)/(3y) =(5(y-1))/(3y)$

Therefore

$a=((10+11y)/(3y))/((5(y-1))/(3y)) =(10+11y)/(5(y-1))$

Answer: dfrac{10+11y}{5(y-1)}

26 Here is a prism $ABCDEFGH$ with a horizontal, rectangular base $ABGF$

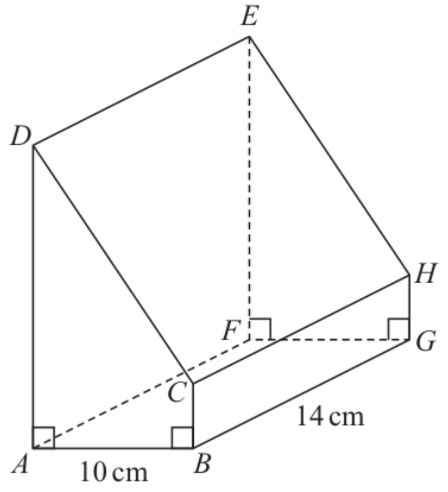


Diagram **NOT** accurately drawn

$AB = 10\text{ cm}$ $\quad$ $BG = 14\text{ cm}$

angle DAB = angle ABC = angle EFG = angle FGH = 90°

$BC = \frac{1}{5}AD$

The angle of elevation of E from B is 50°

Calculate the volume of the prism.
Give your answer correct to 3 significant figures.
Show your working clearly.

.... cm^3

(Total for Question 26 is 5 marks)

ANSWER / WORKED SOLUTION

1H • Family: Resolve multi-stage solid geometry • Variant: Use 3D trigonometry to obtain a terminal volume • USER-ATTESTED LEGACY SOLUTION

Topic check: Corrected from Right-Angled Triangles to 3D Pythagoras & Trigonometry.

Method

The horizontal base ABGF is a rectangle.

$AB=10,\qquad BG=14$

So the horizontal distance from B to F is

$BF=\sqrt{10^2+14^2}=\sqrt{296}$

The angle of elevation of E from B is 50°.

Since E is vertically above F,

$\tan 50^\circ=\frac{FE}{BF}$

$FE=\sqrt{296}\tan50^\circ$

The prism has the same cross section throughout, so

$AD=FE=\sqrt{296}\tan50^\circ$

Also

$BC=\frac15AD$

Area of the trapezium cross section ABCD:

$\frac12(AD+BC)\times AB$

$=\frac12(AD+\frac15AD)\times 10$

$=6AD$

Volume of the prism:

$6AD\times 14=84AD$

$=84\sqrt{296}\tan50^\circ$

$=1722.311\ldots$

Correct to 3 significant figures:

1720

Answer: $\boxed{1720\text{ cm}^3}$

1 The point P has coordinates (3, 4)
The point Q has coordinates (9, 16)

M is the midpoint of the line PQ

Find the coordinates of M

(.......................... , ...)

(Total for Question 1 is 2 marks)

ANSWER / WORKED SOLUTION

2H • Family: Determine coordinates from geometric information • Variant: Rectangle or parallelogram missing vertex • USER-ATTESTED LEGACY SOLUTION

Topic check: Coordinate geometry. The tag is correct.

Solution

P=(-3,0),qquad Q=(9,16)

The midpoint is

((-3+9)/(2),(0+16)/(2))

=(3,8)

Answer: (3,8).

3 A box contains only

9 red bricks
43 blue bricks
and some yellow bricks

$\frac{7}{20}$ of the bricks are yellow bricks.

Each brick weighs 35 grams.

Work out the total weight of the yellow bricks.

.... grams

(Total for Question 3 is 4 marks)

ANSWER / WORKED SOLUTION

2H • Family: Represent, simplify and compare fractions • Variant: Fraction of a quantity or one number as a fraction of another • USER-ATTESTED LEGACY SOLUTION

Topic check: Fractions. The tag is correct.

Method

There are $9+43=52$ red and blue bricks.

Let the total number of bricks be T.

Since $\frac{7}{20}$ of the bricks are yellow, the remaining fraction is

$1-\frac{7}{20}=\frac{13}{20}$

So

$\frac{13}{20}T=52$

$T=52\times \frac{20}{13}=80$

Yellow bricks:

$\frac{7}{20}\times 80=28$

Each brick weighs 35 grams, so total yellow-brick weight is

$28\times 35=980$

Answer: 980 grams

6 A large circle with centre O contains 3 identical small circles.
Each circle touches two other circles.

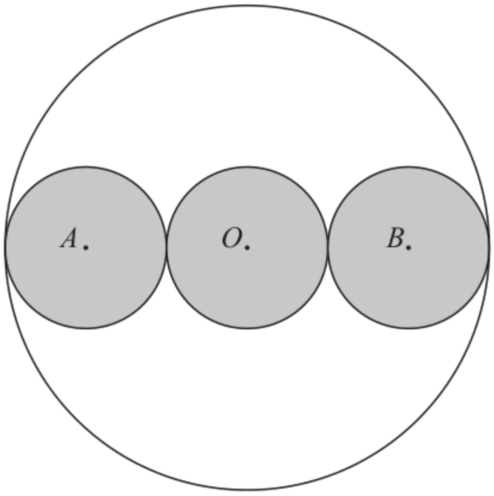


Diagram **NOT** accurately drawn

A, O and B are the centres of the small circles.
AOB is a straight line.

The circumference of the large circle is 160 cm

Work out the area of one small circle.
Give your answer correct to 3 significant figures.

.... cm^2

(Total for Question 6 is 5 marks)

ANSWER / WORKED SOLUTION

2H • Family: Resolve compound and shaded circle regions • Variant: Find a mixed straight-and-curved perimeter • USER-ATTESTED LEGACY SOLUTION

Topic check: Corrected to circles, arcs and sectors.

Solution

The circumference of the large circle is 160 cm, so

$2\pi R=160$

$R=\frac{80}{\pi}$

The three equal small circles fit along the diameter of the large circle, so

$2R=6r$

$r=\frac{R}{3}=\frac{80}{3\pi}$

Area of one small circle is

$\pi r^2=\pi\left(\frac{80}{3\pi}\right)^2$

$=226.35\ldots$

Answer: 226 cm^2 to 3 significant figures.

7 (i) Factorise $x^2 + 2x - 48$

....

(2)

(ii) Hence, solve $x^2 + 2x - 48 = 0$

....

(1)

(Total for Question 7 is 3 marks)

ANSWER / WORKED SOLUTION

2H • Family: Factorise a non-monic or special quadratic • Variant: Factorise as the first part of a factor-then-solve chain • USER-ATTESTED LEGACY SOLUTION

Topic check: Factorising. The tag is correct.

Solution

$x^2+2x-48=(x+8)(x-6)$

So

$(x+8)(x-6)=0$

Answer: $x=-8$ or $x=6$

9 Mark has three bags, bag A, bag B and bag C

Bag A contains 120 counters.
The probability of taking at random a red counter from bag A is 0.45

Bag B contains 80 counters.
The probability of taking at random a red counter from bag B is 0.3

Bag C is empty.

Mark puts all the counters from bag A and bag B into bag C

Mark takes at random a counter from bag C

Work out the probability that he takes a red counter.

....

(Total for Question 9 is 3 marks)

ANSWER / WORKED SOLUTION

2H • Family: Calculate theoretical single-event probability • Variant: Recover population composition from a single-event probability • USER-ATTESTED LEGACY SOLUTION

Topic check: Probability Toolkit. The tag is correct.

Solution

Bag A has

$120\times 0.45=54$

red counters.

Bag B has

$80\times 0.30=24$

red counters.

After combining the bags, the total number of red counters is

$54+24=78$

The total number of counters is

$120+80=200$

$P(\text{red})=\frac{78}{200}=0.39$

Answer: 0.39.

11 The diagram shows a right-angled triangle *ABC*

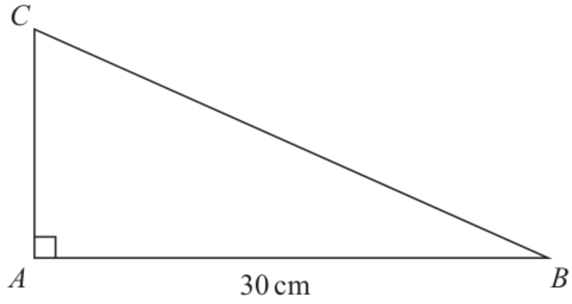


Diagram **NOT** accurately drawn

Area of triangle $ABC = 240\,\text{cm}^2$

Work out the perimeter of triangle *ABC*

.... cm

(Total for Question 11 is 4 marks)

ANSWER / WORKED SOLUTION

2H • Family: Use angles of elevation and depression • Variant: Find a height from an elevation or depression angle • USER-ATTESTED LEGACY SOLUTION

Topic check: Right-Angled Triangles - Pythagoras & Trigonometry. The tag is correct.

Solution

Use the area to find AC:

frac12× 30× AC=240

15AC=240

AC=16

Then

BC^2=30^2+16^2=1156

BC=34

Perimeter:

30+16+34=80

Answer: 80 cm.

13 (a) Simplify fully $\left(\dfrac{125x^6}{y^{15}}\right)^{\frac{2}{3}}$

....

(2)

Given that $4^n = \dfrac{4^m}{64^p}$

(b) Express n in terms of m and p

$n =$ ....

(2)

(Total for Question 13 is 4 marks)

ANSWER / WORKED SOLUTION

2H • Family: Solve an index relation by a common base • Variant: Find a numerical unknown exponent •
USER-ATTESTED LEGACY SOLUTION

Topic check: Algebraic roots and indices. The tag is correct.

Solution

(a)

$\left(\frac{125x^6}{y^{15}}\right)^{1/3} = \frac{5x^2}{y^5}$

(b)

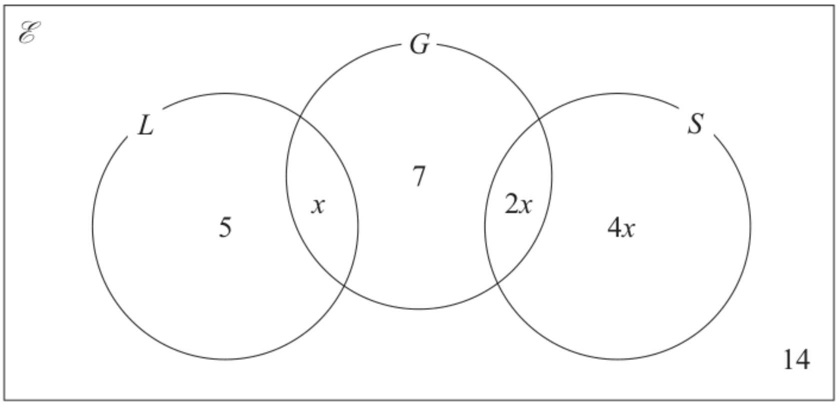
$64^p = (4^3)^p = 4^{3p}$

$4^n = \frac{4^m}{64^p} = \frac{4^m}{4^{3p}} = 4^{m-3p}$

Answer: (a) $\frac{5x^2}{y^5}$, (b) $n = m - 3p$

18 Bianca asked 40 students which of the languages Latin (L), Greek (G) and Sanskrit (S) they study.

The Venn diagram gives some information about her results.
It shows the number of students in each subset.



One of these students is selected at random.
Work out the probability that this student studies Latin or Sanskrit.

....

(Total for Question 18 is 3 marks)

ANSWER / WORKED SOLUTION

2H • Family: Represent and read events with Venn diagrams • Variant: Complete a two-set Venn diagram from totals • USER-ATTESTED LEGACY SOLUTION

Topic check: Probability Diagrams - Venn & Tree Diagrams. The tag is correct.

Solution

The total is 40.

The regions shown add to

5+x+7+2x+4x+14

So

26+7x=40

7x=14

x=2

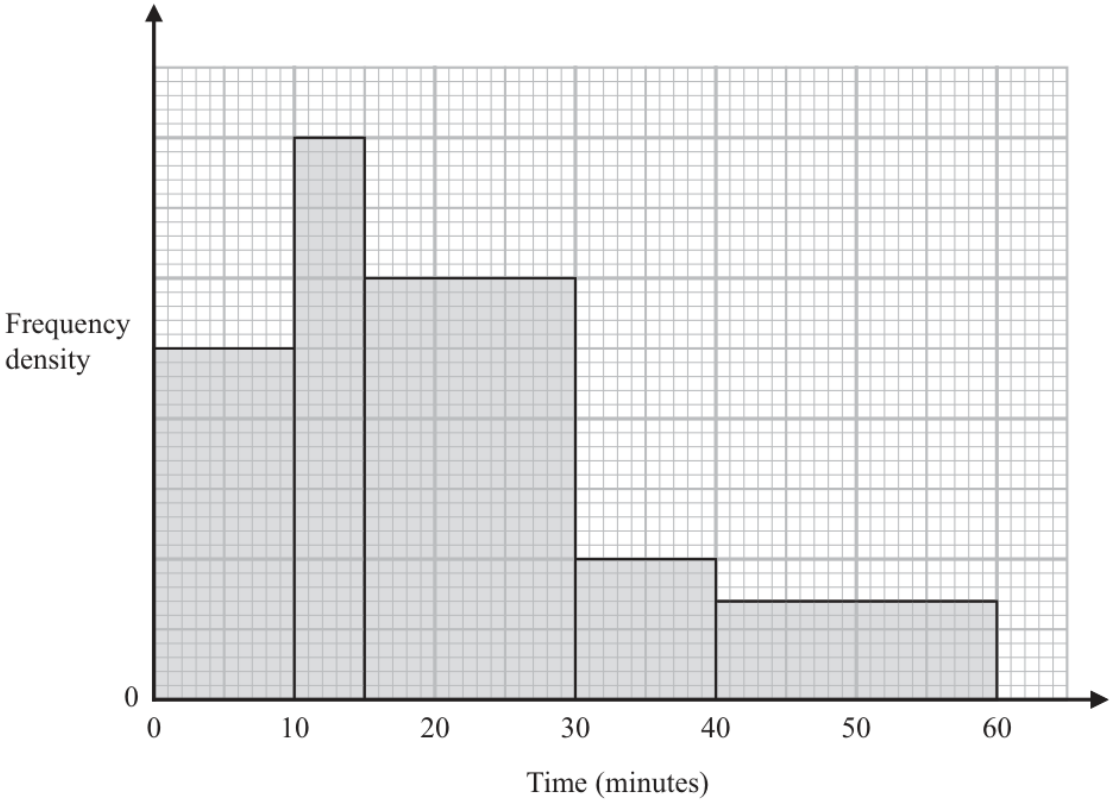
Latin or Sanskrit:

5+x+2x+4x=5+2+4+8=19

P(L□ S)=(19)/(40)

Answer: (19)/(40).

21 The histogram shows information about the times, in minutes, that some trains arrived late at a station one day.



20 of these trains arrived between 10 minutes late and 15 minutes late.

No trains arrived more than 60 minutes late.

Work out an estimate for the number of these trains that arrived at least 25 minutes late.

....

(Total for Question 21 is 3 marks)

ANSWER / WORKED SOLUTION

2H • Family: Interpret histogram areas • Variant: Estimate a histogram proportion, fraction or percentage
• USER-ATTESTED LEGACY SOLUTION

Topic check: Correct.

Method

For a histogram,

frequency=class width× frequency density

For $10<t\le15$, the frequency is 20 and the class width is 5, so

frequency density=(20)/(5)=4

Using this scale from the histogram:

$15<t\le30$ quad has frequency density 3

$30<t\le40$ quad has frequency density 1

$40<t\le60$ quad has frequency density 0.7

For trains at least 25 minutes late:

(5)(3)+(10)(1)+(20)(0.7)

=15+10+14

=39

Answer: 39 trains

24 $D = \dfrac{n}{p - q}$

$n = 10.3$ correct to 1 decimal place
$p = 7.24$ correct to 2 decimal places
$q = 4.39$ correct to 2 decimal places

By considering bounds, work out the value of D to a suitable degree of accuracy.
Show your working clearly.

....

(Total for Question 24 is 5 marks)

ANSWER / WORKED SOLUTION

2H • Family: Round to a stated accuracy • Variant: Round to decimal places or significant figures • USER-ATTESTED LEGACY SOLUTION

Topic check: Rounding, Estimation and Bounds. The tag is correct.

Method

D=(n)/(p-q)

Bounds:

10.25le n<10.35

7.235le p<7.245

4.385le q<4.395

Lower bound:

D_(lower)=(10.25)/(7.245-4.385)=3.5839ldots

Upper bound:

D_(upper)=(10.35)/(7.235-4.395)=3.6443ldots

So

3.5839ldots < D < 3.6443ldots

Both bounds round to 3.6 to 1 decimal place.

Answer: D=3.6 to a suitable degree of accuracy